Important Note: If $m_1 m_2 = -1$, then the two lines are perpendicular. But the converse is not true, because the lines are parallel to the axes, the result cannot be applied, even when the lines are perpendicular.

6.4.3 Position of a point with respect to a straight line

Any line $ax + by + c = 0$ $(c \neq 0)$,divides the whole plane into two parts:

(i) one containing the origin called origin side of the line and
(ii) the other not containing the origin called non-origin side of the line.
A point $P(x_1, y_1)$ is on the origin side or non-origin side of the line $ax + by + c = 0$ $(c \neq 0)$, according as $ax_1 + by_1 + c$ and c are of the same sign or opposite sign.
If $c > 0$, then $P(x_1, y_1)$ is on the origin side or non origin side of the line $ax + by + c = 0$, according as $ax_1 + by_1 + c$ is positive or negative.

After rewriting the equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, such that both $c_1 > 0$ and $c_2 > 0$, and if

(i) $a_1a_2 + b_1b_2 < 0$, then the angle between them is acute
(ii) $a_1a_2 + b_1b_2 > 0$, then the angle between them is obtuse.

6.4.4 Distance Formulas

Let us develop formulas to find the distance between

(i) two points
(ii) a point to a line
(iii) two parallel lines

(i) The distance between two points (x_1, y_1) and (x_2, y_2) is given by the formula

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The above formula was already proved in lower class.

(ii) The distance from a point $P(x_1, y_1)$ to a line $ax + by + c = 0$ is $\left| \dfrac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$

Proof. Let AB be the given line,

$$ax + by + c = 0 \tag{6.20}$$

$P(x_1, y_1)$ be the given point.

Draw a line *CD* parallel to *AB* through P and drop a perpendicular from $P(x_1, y_1)$ to *AB* to meet at M. Also drop a perpendicular from the origin to the line AB to meet at R, and meeting *CD* at Q.

Let $\angle AOR = \alpha$

Therefore the normal form of the the line AB is

$$x \cos\alpha + y \sin\alpha = p \tag{6.21}$$

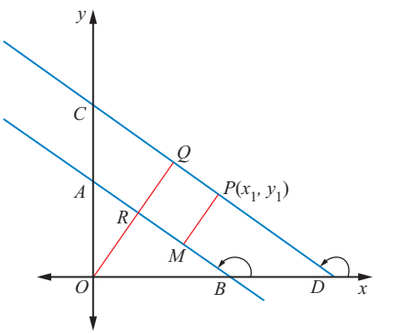


Figure 6.37